

矩阵分析与应用

第十四讲 广义逆矩阵

本讲主要内容

- 投影变换
- 广义逆的存在、性质及构造方法
- 广义逆矩阵的计算方法

投影矩阵

定义：向量空间 C^n 中，子空间 L 与 M 满足 $C^n = L \oplus M$ ，
对 $\forall x \in C^n$ ，分解式 $x = y + z, y \in L, z \in M$ 唯一。

称变换 $T_{L,M}(x) = y$ 为沿着 M 到 L 的投影

性质(1): $T_{L,M}$ 是线性变换

性质(2): $R(T_{L,M}) = L, N(T_{L,M}) = M$

性质(3): $\forall x \in L \Rightarrow T_{L,M}(x) = x \quad \forall x \in M \Rightarrow T_{L,M}(x) = \theta$

[注] $T_{L,M}$ 是 L 中的单位变换

$T_{L,M}$ 是 M 中的零变换

二、投影矩阵

定义：取线性空间 C^n 的基为 $e_1, e_2, \dots, e_n$ 时，元素 x 与它的坐标“形式一致”。称 $T_{L,M}$ 在该基下的矩阵为投影矩阵，记为 $P_{L,M}$

性质(4): $T_{L,M}(x) = y \Leftrightarrow P_{L,M}x = y$

$$x \in L \Rightarrow T_{L,M}(x) = x \Rightarrow P_{L,M}x = x$$

$$x \in M \Rightarrow T_{L,M}(x) = \theta \Rightarrow P_{L,M}x = \theta$$

预备: $R(A) = \{y \mid y = Ax, x \in C^n\}, N(A) = \{x \mid Ax = 0, x \in C^n\}$

引理1: $A_{n \times n}, A^2 = A \Rightarrow N(A) = R(I - A)$

证明: $A^2 = A \Rightarrow A(I - A) = 0$

先证 $R(I - A) \subset N(A)$

$$\forall x \in R(I - A) \Rightarrow \exists u \in C^n, \text{st. } x = (I - A)u$$

$$Ax = A(I - A)u = 0 \Rightarrow x \in N(A)$$

再证 $N(A) \subset R(I - A): \quad \forall \alpha \in N(A) \Rightarrow A\alpha = 0$

$$\alpha = \alpha - A\alpha = (I - A)\alpha \in R(I - A)$$

故 $N(A) = R(I - A)$

定理1: $P_{n \times n} = P_{L,M} \Leftrightarrow P^2 = P$

证明: 必要性 $C^n = L \oplus M$

$$\forall x \in C^n, x = y + z, y \in L, z \in M \text{ 唯一} \Rightarrow P_{L,M} x = y$$

$$P_{L,M}^2 x = P_{L,M} (P_{L,M} x) = P_{L,M} y = y = P_{L,M} x$$

充分性 $\forall x \in C^n \Rightarrow x = Px + (I - P)x$

$$\text{令 } y = Px \in R(P), z = (I - P)x \in R(I - P) = N(P)$$

则 $C^n = R(P) + N(P)$, 下证 $R(P) \cap N(P) = \{\theta\}$

对 $\forall \beta \in R(P) \cap N(P)$ $\beta \in R(P) \Rightarrow \exists u \in C^n, \text{st. } \beta = Pu$

$$\beta \in N(P) \Rightarrow P\beta = \theta$$

$$\text{故 } \beta = Pu = P^2 u = PPu = P\beta = \theta$$

于是可得 $C^n = R(P) \oplus N(P)$, 从而有

$$x = y + z, y = Px \in R(P), z = (I - P)x \in N(P) \quad \text{唯一}$$

因为投影变换 $T_{R(P), N(P)}$ 满足

$$T_{R(P), N(P)}(x) = y \Rightarrow P_{R(P), N(P)}x = y \quad (\forall x \in C^n)$$

所以 $P = P_{R(P), N(P)}$

三、投影矩阵的确定方法

$\dim L = r, L$ 的基为 $x_1, \dots, x_r : X = (x_1, \dots, x_r)$

$\dim M = n - r, M$ 的基为 $y_1, \dots, y_{n-r} : Y = (y_1, \dots, y_{n-r})$

$$\left. \begin{array}{l} P_{L,M} x_i = x_i \Rightarrow P_{L,M} X = X \\ P_{L,M} y_j = \theta \Rightarrow P_{L,M} Y = O \end{array} \right\} \Rightarrow P_{L,M} (X | Y) = (X | O)$$

$$\Rightarrow P_{L,M} = (X | O)(X | Y)^{-1}$$

例1: $\mathbf{R}^2$ 中: $\alpha_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, L = L(\alpha_1), M = L(\alpha_2),$

求 $P_{L,M}$

$$\text{解: } P_{L,M} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\text{验证: } P_{L,M} \alpha_1 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$P_{L,M} \alpha_2 = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

例2: $P_{L,M}$ 与 L 和 M 的基的选择无关。

证: L 的基 $x_1, \dots, x_r$; 另一基 $\tilde{x}_1, \dots, \tilde{x}_r$

$$X = (x_1, \dots, x_r), \quad \tilde{X} = (\tilde{x}_1, \dots, \tilde{x}_r) \Rightarrow \tilde{X} = XC_{r \times r}$$

M 的基 $y_1, \dots, y_{n-r}$; 另一基 $\tilde{y}_1, \dots, \tilde{y}_{n-r}$:

$$Y = (y_1, \dots, y_{n-r}), \quad \tilde{Y} = (\tilde{y}_1, \dots, \tilde{y}_{n-r}) \Rightarrow \tilde{Y} = YD_{(n-r) \times (n-r)}$$

$$\begin{aligned} (\tilde{X} | O) \cdot (\tilde{X} | \tilde{Y})^{-1} &= (XC | O) \cdot \left[(X | Y) \begin{pmatrix} C & O \\ O & D \end{pmatrix} \right]^{-1} \\ &= (XC | O) \cdot \begin{bmatrix} C^{-1} & O \\ O & D^{-1} \end{bmatrix} \cdot (X | Y)^{-1} = (X | O) \cdot (X | Y)^{-1} \end{aligned}$$

四、正交投影变换

欧氏空间 C^n 中，子空间 L 给定，取 $M = L^\perp$ ，
则 $C^n = L \oplus M$

正交投影变换 $T_L = T_{L,M}$ ；正交投影矩阵 $P_L = P_{L,M}$

定理2： 方阵 $P = P_L \Leftrightarrow P^2 = P, P^H = P$

证明： 必要性： 由 $P = P_L \Rightarrow P^2 = P$

$$\left. \begin{aligned} \forall x_1 \in C^n, x_1 = y_1 + z_1, y_1 \in L, z_1 \in M \\ \forall x_2 \in C^n, x_2 = y_2 + z_2, y_2 \in L, z_2 \in M \end{aligned} \right\}$$

$$\left. \begin{aligned} P_L x_1 = y_1 \in L, (I - P_L) x_1 = x_1 - y_1 = z_1 \in L^\perp \\ P_L x_2 = y_2 \in L, (I - P_L) x_2 = x_2 - y_2 = z_2 \in L^\perp \end{aligned} \right\}$$

定理2: 方阵 $P = P_L$

$$P_L x_1 = y_1 \in L, (I - P_L) x_1 = x_1 - y_1 = z_1 \in L^\perp$$

$$P_L x_2 = y_2 \in L, (I - P_L) x_2 = x_2 - y_2 = z_2 \in L^\perp$$

$$P_L x_1 \perp (I - P_L) x_2 \Rightarrow x_1^H P_L^H (I - P_L) x_2 = 0$$

$$(I - P_L) x_1 \perp P_L x_2 \Rightarrow x_1^H (I - P_L)^H P_L x_2 = 0$$

$$\text{因此 } x_1^H (P_L^H - P_L) x_2 = 0 \Rightarrow P_L^H - P_L = 0 : P_L^H = P_L$$

充分性: 已知 $P^2 = P \Rightarrow P = P_{R(P), N(P)}$

$$P^H = P : N(P) = N(P^H) = R^\perp(P)$$

$$\text{因此 } x_1^H P = P_{R(P)}$$

四、正交投影矩阵的确定方法

$$\left. \begin{array}{l} L \text{ 的基为 } x_1, \dots, x_r : X = (x_1, \dots, x_r) \\ L^\perp \text{ 的基为 } y_1, \dots, y_{n-r} : Y = (y_1, \dots, y_{n-r}) \end{array} \right\} \Rightarrow \begin{cases} X^H Y = O \\ Y^H X = O \end{cases}$$

$$\text{已求得 } P_L = P_{L, L^\perp} = (X | O) \cdot (X | Y)^{-1}$$

$$\text{因为 } (X | Y)^H \cdot (X | Y) = \begin{pmatrix} X^H \\ Y^H \end{pmatrix} \cdot (X | Y) = \begin{bmatrix} X^H X & O \\ O & Y^H Y \end{bmatrix}$$

$$\text{所以 } (X | Y)^{-1} = \begin{bmatrix} (X^H X)^{-1} & O \\ O & (Y^H Y)^{-1} \end{bmatrix} \cdot (X | Y)^H = \begin{bmatrix} (X^H X)^{-1} X^H \\ (Y^H Y)^{-1} Y^H \end{bmatrix}$$

$$\text{于是 } P_L = (X | O) \cdot \begin{bmatrix} (X^H X)^{-1} X^H \\ (Y^H Y)^{-1} Y^H \end{bmatrix} = X \cdot (X^H X)^{-1} \cdot X^H$$

例3: 向量空间 $\mathbf{R}^3$ 中, $\alpha = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \beta = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, L = L(\alpha, \beta),$

求 P_L

解: $X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}, X^T X = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}, (X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$

$$P_L = X (X^T X)^{-1} X^T = \frac{1}{6} \begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & 1 \\ -2 & 1 & 5 \end{bmatrix}$$

[注]: 正交投影矩阵 P_L 与子空间 L 的基的选择无关

广义逆矩阵

一、定义与算法

定义：对 $A_{m \times n}$ ，若有 $X_{n \times m}$ 满足Penrose方程

$$(1) \quad AXA = A$$

$$(2) \quad XAX = X$$

$$(3) \quad (AX)^H = AX$$

$$(4) \quad (XA)^H = XA$$

称 X 为 A 的M-P逆，记作 A^+ .(Moore 1920, Penrose1955)

例如 $A_{n \times n}$ 可逆, $X = A^{-1}$ 满足P-方程: $A^+ = A^{-1}$

$$A = O_{m \times n}, X = O_{n \times m}$$

满足P-方程: $O_{m \times n}^+ = O_{n \times m}$

$$A = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, X = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

满足P-方程: $A^+ = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \end{bmatrix}$

例4: $F \in \mathbf{C}_r^{m \times r}$ ($r \geq 1$) $\Rightarrow F^+ = (F^H F)^{-1} F^H$, 且 $F^+ F = I_r$

$G \in \mathbf{C}_r^{r \times n}$ ($r \geq 1$) $\Rightarrow G^+ = G^H (G G^H)^{-1}$, 且 $G G^+ = I_r$

验证第一式: 令 $F^+ = (F^H F)^{-1} F^H$, 则有

$$F X F = F (F^H F)^{-1} F^H F = F$$

$$X F X = (F^H F)^{-1} F^H F X = X$$

$$(F X)^H = X^H F^H = F (F^H F)^{-1} F^H = F X$$

$$(X F)^H = I_r^H = I_r = X F$$

定理3: $\forall A_{m \times n}, A^+$ 存在并唯一

证明: 存在性 $A = O_{m \times n} \Rightarrow A^+ = O_{n \times m}$

$$A \neq O \Rightarrow \text{rank} A \geq 1: A = FG, F \in \mathbf{C}_r^{m \times r}, G \in \mathbf{C}_r^{r \times n}$$

令 $X = G^+ F^+$ 则有

$$AXA = FG \cdot G^+ F^+ \cdot FG = FG = A$$

$$XAX = G^+ F^+ \cdot FG \cdot G^+ F^+ = G^+ F^+ = X$$

$$(AX)^H = (FG \cdot G^+ F^+)^H = (FF^+)^H = FF^+ = F \cdot GG^+ \cdot F^+ = AX$$

$$(XA)^H = (G^+ F^+ \cdot FG)^H = (G^+ G)^H = G^+ G = G^+ \cdot F^+ F \cdot G = XA$$

$$A^+ = G^+ F^+ = G^H (F^H A G^H)^{-1} F^H$$

定理3: $\forall A_{m \times n}, A^+$ 存在并唯一

证明：唯一性，对 $A_{m \times n}$ 若 $X_{n \times m}$ 与 $Y_{n \times m}$ 都满足P-方程，
则：

$$\begin{aligned} X &= XAX = X \cdot AYA \cdot X = X \cdot (AY)^H \cdot (AX)^H \\ &= X \cdot (AXAY)^H = X \cdot (AY)^H = XAY = X \cdot AYA \cdot Y \\ &= (XA)^H \cdot (YA)^H \cdot Y = (YAXA)^H \cdot Y = (YA)^H \cdot Y = YAY = Y \end{aligned}$$

例5： 设 $A \in \mathbb{C}_r^{m \times n}$ 的奇异值分解为 $A = U \begin{bmatrix} \Sigma_r & O \\ O & O \end{bmatrix}_{m \times n} V^H$

$$\text{则 } A^+ = V \begin{bmatrix} \Sigma_r^{-1} & O \\ O & O \end{bmatrix}_{n \times m} U^H$$

直接验证即可。

进一步的有

$$A = (U_s, U_n) \begin{bmatrix} \Sigma_r & O \\ O & O \end{bmatrix}_{m \times n} (V_s, V_n)^H$$

$$= U_s \Sigma_r V_s^H$$

$$A^+ = V_s \Sigma_r^{-1} U_s^H$$

广义逆矩阵的分类：对 $A_{m \times n}$ ，若 $X_{n \times m}$ 满足 **P**-方程

(i): 称 X 为 A 的 $\{i\}$ -逆，记作 $A^{(i)}$ 。全体记作 $A\{i\}$

(i),(j): 称 X 为 A 的 $\{i,j\}$ -逆，记作 $A^{(i,j)}$ 。全体记作 $A\{i,j\}$

(i),(j),(k): 称 X 为 A 的 $\{i,j,k\}$ -逆，记作 $A^{(i,j,k)}$ 。全体记作 $A\{i,j,k\}$

(1)~(4): 则 X 为 A^+

合计：15类

常用广义逆矩阵： $A\{1\}, A\{1,2\}, A\{1,3\}, A\{1,4\}, A^+$

求 $A^{(1)}, A^{(1,2)}$ 的初等变换方法

$$A \in \mathbf{C}_r^{m \times n}, \quad A \xrightarrow{\text{行}} B \Rightarrow \exists \text{可逆矩阵 } Q_{m \times m}, \text{ st. } QA = B$$

其中 B 为拟Hermite标准形, 它的后 $m - r$ 行元素全为零

$$B \xrightarrow{\text{列对换}} \begin{bmatrix} I_r & K \\ O & O \end{bmatrix}_{m \times n} = C \Rightarrow \exists \text{置换矩阵 } P_{n \times n}, \text{ st. } BP = C$$

于是 $QAP = C \Rightarrow A = Q^{-1}CP^{-1}$

定理14: 已知 A , P , Q 如上所述, 对 $\forall L_{(n-r) \times (m-r)}$, 有

$$X = P \begin{bmatrix} I_r & O \\ O & L \end{bmatrix}_{n \times m} Q \in A\{1\}, \quad X_0 = P \begin{bmatrix} I_r & O \\ O & O \end{bmatrix}_{n \times m} Q \in A\{1,2\}$$

证明: $AXA = Q^{-1}CP^{-1} \cdot P \begin{bmatrix} I_r & 0 \\ 0 & L \end{bmatrix}_{n \times m} Q \cdot Q^{-1}CP^{-1}$

$$= Q^{-1} \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix}_{m \times n} \begin{bmatrix} I_r & 0 \\ 0 & L \end{bmatrix}_{n \times m} \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix}_{m \times n} P^{-1}$$

$$= Q^{-1} \begin{bmatrix} I_r & KL \\ 0 & 0 \end{bmatrix}_{m \times m} \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix}_{m \times n} P^{-1}$$

$$= Q^{-1} \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix}_{m \times n} P^{-1} = A \quad \text{故} \quad X \in A\{1\}$$

显然 $AX_0A = A$, 且有

$$\begin{aligned} X_0AX_0 &= P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q \cdot Q^{-1} \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix} P^{-1} \cdot P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q \\ &= P \begin{bmatrix} I_r & K \\ 0 & 0 \end{bmatrix} \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q \\ &= P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q = X_0 \end{aligned}$$

故 $X_0 \in A\{1,2\}$

例6: $A = \begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & 1 & 4 \end{bmatrix}$, 求 $A^{(1)}, A^{(1,2)}, A^+$

解: $(A | I) \rightarrow \left[\begin{array}{cccc|ccc} 2 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array} \right] : c_1 = 2, c_2 = 3$

$$Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}, \quad P = (e_2, e_3, e_1, e_4) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A^{(1)} = P \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & a \\ 0 & 0 & b \end{array} \right] Q = \begin{bmatrix} -a & -a & a \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & -b & b \end{bmatrix}, \quad A^{(1,2)} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = FG :$$

$$F = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, F^T F = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, F^+ = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 2 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix}, GG^T = \begin{bmatrix} 9 & 4 \\ 4 & 5 \end{bmatrix}, G^+ = \frac{1}{29} \begin{bmatrix} 10 & -8 \\ 5 & -4 \\ -4 & 9 \\ 2 & 10 \end{bmatrix}$$

$$A^+ = G^+ F^+ = \frac{1}{87} \begin{bmatrix} 28 & -26 & 2 \\ 14 & -13 & 1 \\ -17 & 22 & 5 \\ -6 & 18 & 12 \end{bmatrix}$$

例7 : $A_{m \times n} \neq O$, 且 A^+ 已知, 记 $B = \begin{bmatrix} A \\ A \end{bmatrix}$, 求 B^+

解: $\text{rank} A = r \geq 1 \Rightarrow A = FG : F \in \mathbf{C}_r^{m \times r}, G \in \mathbf{C}_r^{r \times n}$

$$B = \begin{pmatrix} FG \\ FG \end{pmatrix} = \begin{pmatrix} F \\ F \end{pmatrix} G : \begin{pmatrix} F \\ F \end{pmatrix} \in \mathbf{C}_r^{2m \times r}, G \in \mathbf{C}_r^{r \times n}$$

$$B^+ = G^+ \begin{pmatrix} F \\ F \end{pmatrix}^+ = G^+ \cdot \left[\begin{pmatrix} F^H & | & F^H \end{pmatrix} \begin{pmatrix} F \\ F \end{pmatrix} \right]^{-1} \begin{pmatrix} F^H & | & F^H \end{pmatrix}$$

$$= G^+ \cdot \frac{1}{2} (F^H F)^{-1} \cdot \begin{pmatrix} F^H & | & F^H \end{pmatrix} = \frac{1}{2} (G^+ F^+ | G^+ F^+) = \frac{1}{2} (A^+ | A^+)$$

二、广义逆矩阵的性质

定理4: $A_{m \times n}, A^{(1)}$ 唯一 $\Leftrightarrow m = n, A$ 可逆, 且 $A^{(1)} = A^{-1}$

定理5: $A_{m \times n}, B_{n \times p}, \lambda \in \mathbf{C}, \lambda^+ = \begin{cases} 1/\lambda & (\lambda \neq 0) \\ 0 & (\lambda = 0) \end{cases}$

$$(1) \quad [A^{(1)}]^H \in A^H \{1\}: AA^{(1)}A = A \Rightarrow A^H (A^{(1)})^H A^H = A^H$$

$$(2) \quad \lambda^+ A^{(1)} \in (\lambda A) \{1\}: (\lambda A)(\lambda^+ A^{(1)})(\lambda A) = (\lambda \lambda^+ \lambda)(AA^{(1)}A) = \lambda A$$

$$(3) \quad S_{m \times m} \text{ 和 } T_{n \times n} \text{ 都可逆} \Rightarrow T^{-1} A^{(1)} S^{-1} \in (SAT) \{1\}$$

$$(4) \quad r_A \leq r_{A^{(1)}}: r_A = r_{AA^{(1)}A} \leq r_{A^{(1)}}$$

(5) $AA^{(1)}$ 与 $A^{(1)}A$ 都是幂等矩阵, 且 $r_{AA^{(1)}} = r_A = r_{A^{(1)}A}$

$$\text{因为 } r_A = r_{AA^{(1)}A} \leq \begin{cases} r_{AA^{(1)}} \\ r_{A^{(1)}A} \end{cases} \leq r_A$$

$$(6) \quad R(AA^{(1)}) = R(A): \quad R(A) = R(AA^{(1)}A) \subset R(AA^{(1)}) \subset R(A)$$

$$N(A^{(1)}A) = N(A): \quad N(A) \subset N(A^{(1)}A) \subset N(AA^{(1)}A) = N(A)$$

$$(7) \quad \textcircled{1} \quad A^{(1)}A = I_n \Leftrightarrow r_A = n \quad \text{“}A\text{ 列满秩”}$$

$$\textcircled{2} \quad AA^{(1)} = I_m \Leftrightarrow r_A = m \quad \text{“}A\text{ 行满秩”}$$

$$(8) \quad \textcircled{1} \quad (AB)(AB)^{(1)}A = A \Leftrightarrow r_{AB} = r_A$$

$$\textcircled{2} \quad B(AB)^{(1)}(AB) = B \Leftrightarrow r_{AB} = r_B$$

定理6: $A_{m \times n}, Y \in A\{1\}, Z \in A\{1\} \Rightarrow X \triangleq YAZ \in A\{1,2\}$

证明: $AXA = A \cdot YAZ \cdot A = AY(AZA) = AYA = A$

$$XAX = YAZ \cdot A \cdot YAZ = Y(AZA)YAZ = Y \cdot AYA \cdot Z = YAZ = X$$

推论 $A_{m \times n}, Y \in A\{1\} \Rightarrow X \triangleq YAY \in A\{1,2\}$

定理7: 设 $X \in A\{1\}$, 则 $r_X = r_A \Leftrightarrow X \in A\{1,2\}$

定理8: $Y \triangleq (A^H A)^{(1)} A^H \in A\{1,2,3\}, Z \triangleq A^H (A A^H)^{(1)} \in A\{1,2,4\}$

定理9: $A^+ = A^{(1,4)} A A^{(1,3)}$

定理10: (1) $r_{A^+} = r_A$ (2) $(A^+)^+ = A$

$$(3) \quad (A^H)^+ = (A^+)^H \quad (A^T)^+ = (A^+)^T$$

$$(4) \quad (A^H A)^+ = A^+ (A^H)^+ \quad (A A^H)^+ = (A^H)^+ A^+$$

$$(5) \quad A^+ = (A^H A)^+ A^H = A^H (A A^H)^+$$

$$(6) \quad R(A^+) = R(A^H), N(A^+) = N(A^H)$$

二、M-P逆的等价定义

Moore逆: 对 $A_{m \times n}$, 若有 $X_{n \times m}$ 满足 $AX = P_{R(A)}$

和 $XA = P_{R(X)}$, 称 X 为 A 的 Moore 逆。

定理11: M-逆与P-逆等价

证明: (1) 设 X 是 A 的 M-逆:

$$AXA = P_{R(A)} \cdot (a_1, \dots, a_n) = (a_1, \dots, a_n) = A$$

$$XAX = P_{R(X)} \cdot (x_1, \dots, x_m) = (x_1, \dots, x_m) = X$$

$$(AX)^H = P_{R(A)}^H = P_{R(A)} = AX$$

$$(XA)^H = P_{R(X)}^H = P_{R(X)} = XA \quad \text{故 } X \text{ 是 } A \text{ 的 P-逆}$$

(2) 设 X 是 A 的 P-逆:

$$\left. \begin{array}{l} (AX)^2 = AXAX = AX \\ (AX)^H = AX \end{array} \right\} \Rightarrow AX = P_{R(AX)} = P_{R(A)}$$

$$\because AXA = A, R(AXA) \subset R(AX) \subset R(A)$$

$$\left. \begin{array}{l} (XA)^2 = XAXA = XA \\ (XA)^H = XA \end{array} \right\} \Rightarrow XA = P_{R(XA)} = P_{R(X)}$$

$$\because XAX = A, R(XAX) \subset R(XA) \subset R(X)$$

故 X 是 A 的 M-逆

作业

- P296: 1、2、3、5、7
- P306: 7、8、9